

Retail Store Sales Analysis — Auto Analysis

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Executive Summary

This report summarizes the dataset quality, exploratory findings, and model performance for project "Retail Store Sales Analysis".

Dataset Overview

File	retail_sales_dataset.csv
Rows	3000
Columns	14
Missing Values	0
Duplicate Rows	0
Cleaned	True

Modeling & Evaluation

Model Type	decision_tree
Problem Type	classification
Target Column	store_city

Metrics

Metric	Value
accuracy	0.2483
precision	0.2475
recall	0.2483
f1	0.2479
roc_auc	0.5136

confusion_matrix	{'labels': ['Faisalabad', 'Islamabad', 'Karachi', 'Lahore', 'Multan'], 'matrix': [[14, 18, 25, 19, 6], [14, 28, 46, 28, 10], [23, 35, 54, 50, 16], [15, 34, 36, 46, 22], [12, 10, 17, 15, 7]]}
class_distribution	{'Karachi': 178, 'Lahore': 153, 'Islamabad': 126, 'Faisalabad': 82, 'Multan': 61}
cv_scores	[0.22190675666634466, 0.20871921252752212, 0.2012526080864572, 0.21382144324952654, 0.22690754552602085]
cv_mean	0.2145

Feature Importance

Feature	Importance
profit	0.185
sales_amount	0.179
unit_price	0.1749
customer_rating	0.1178
quantity	0.0619
discount_percent	0.0574
customer_segment_New	0.0309
sales_channel_In-Store	0.0278
sales_channel_Mobile App	0.0196
promotion_applied_No	0.0166
customer_segment_Regular	0.0164
customer_segment_Premium	0.0159
sales_channel_Online	0.0143
product_category_Beauty	0.0126
returned_Yes	0.0124
promotion_applied_Yes	0.0124
returned_No	0.0119

product_category_Home & Kitchen	0.0114
product_category_Sports	0.0098
product_category_Clothing	0.0071

AI Business Insight AI-generated insight

AI service is currently unavailable. This report contains verified analytical results only; AI-generated narrative could not be produced.

Limitations

Model predictions are estimates based on historical data and may contain uncertainty. AI-generated text explains verified analytical results but does not independently calculate metrics. Always validate significant business decisions with domain expertise.